GE VERNOVA

GridOS SCADA

0.7.0

User Guide

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Change History

Date	Change Description
2025-Apr-03	GridOS SCADA 0.1.0: • This is a new document for the initial release of GridOS SCADA.
2025-Jun-24	GridOS SCADA 0.3.0: • Added Controls, which describes how to issue controls in the Substation Tabular display. • Added SCADA Control Logging and Event Flow.
2025-Aug-22	GridOS SCADA 0.4.0: • Added SCADA Control Permissions and AORs Validation to provide guidance on how user permissions and AOR validation work for Control operations. • Updated SCADA Controls with details about issuing control commands for multiphase points.
2026-Mar-03	GridOS SCADA 0.7.0: • Added Site Communication Status to provide details about the communication status of sites. • Editorial and formatting updates.

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1. Introduction

This document describes the procedures available to Supervisory Control and Data Acquisition (SCADA) operators for performing grid operation functions using GridOS SCADA applications.

GridOS SCADA applications are a set of highly scalable microservices operating on the GridOS platform that provide telemetered values in the network and warn users about abnormal conditions in the network.

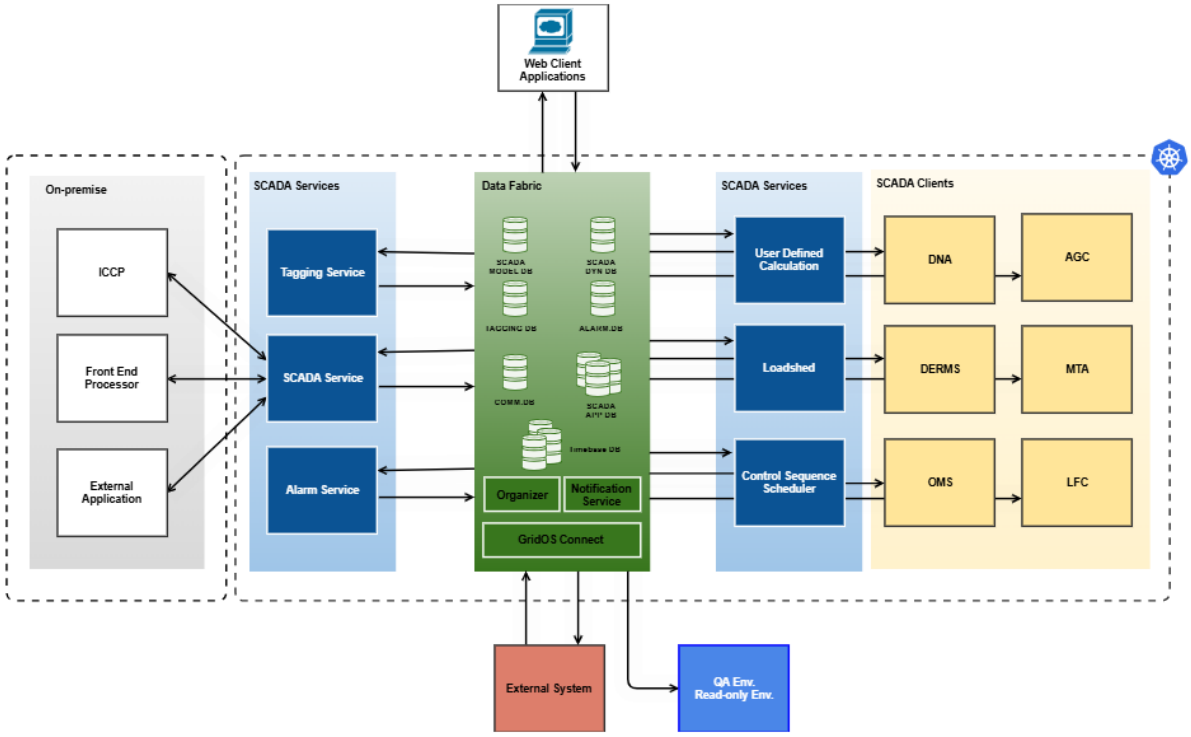
The GridOS SCADA applications provide the following functions:

- **User-Defined Calculation:** The User-Defined Calculation application performs operator-defined general calculations.
- **Controls:** Issues control to Heritage SCADA when configured in hybrid mode.
- **SCADA Data Replication**: Replicates SCADA data from Heritage SCADA when configured in hybrid mode.

1.1. GridOS SCADA Applications System Overview

GridOS SCADA applications rely on the underlying GridOS platform services for logging, monitoring, and user management.

The following system overview depicts the GridOS SCADA applications runtime environment:



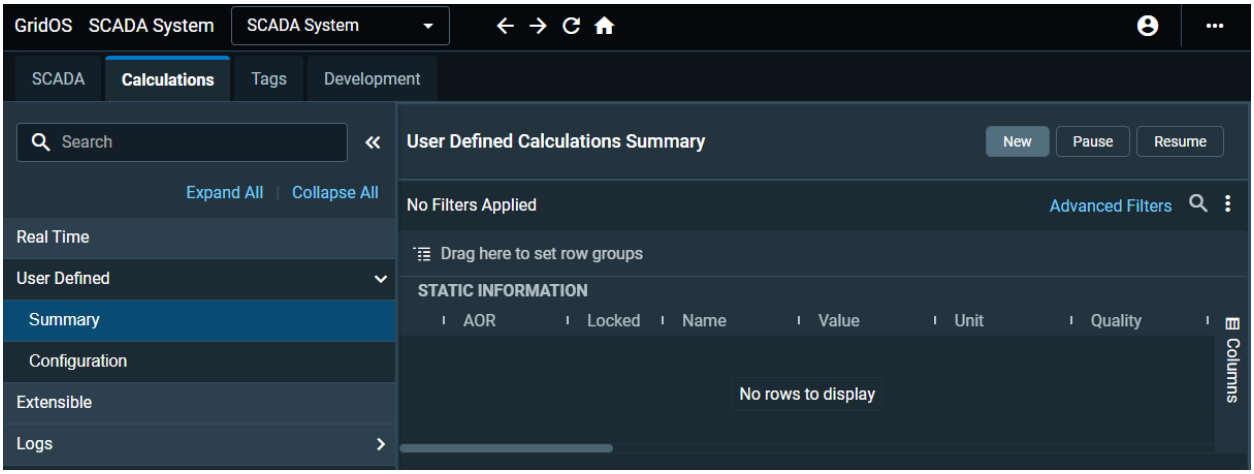
1.2. GridOS SCADA Applications Related Resources

For more information about GridOS SCADA applications deployment and configurations, refer to the following documents:

- *GridOS SCADA Installation Guide*

2. User-Defined Calculations

This chapter covers adding, modifying, deleting, and copying operator-defined calculations. Each of these tasks is performed from the **User Defined Calculations Summary** display.



2.1. Adding a Calculation

This section describes the procedure for inserting a new calculation, which might include any combination of valid logical, boolean, and arithmetic operations available to the user (for details about valid operations, see Additional User Calculation Information).

To add a new calculation:

1. Open the **User Defined Calculations Summary** display.

   The User Defined Calculations Summary appears, listing the existing user-defined calculations.

2. Insert a new calculation using one of the following methods:

   First method:

   - Click **New** to insert a default calculation into the list.

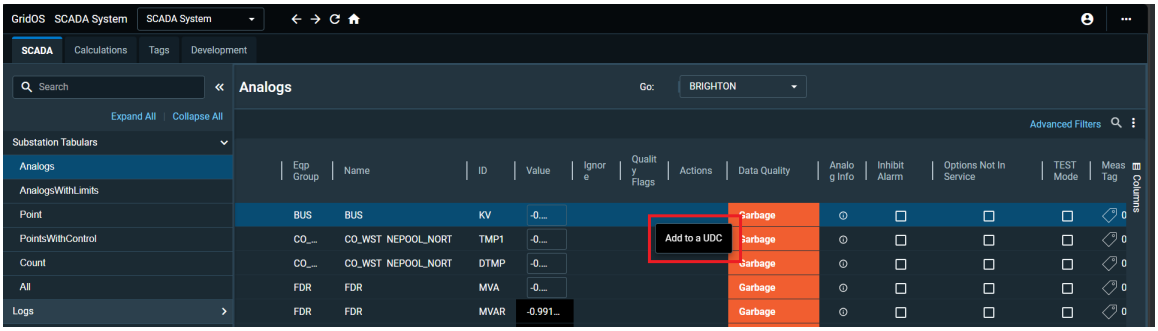
   Second Method:

   - Open the **SCADA Substation Tabulars Analogs** display that is delivered with the GridOS SCADA release, and right-click the **Actions** column.

     The Actions menu appears.

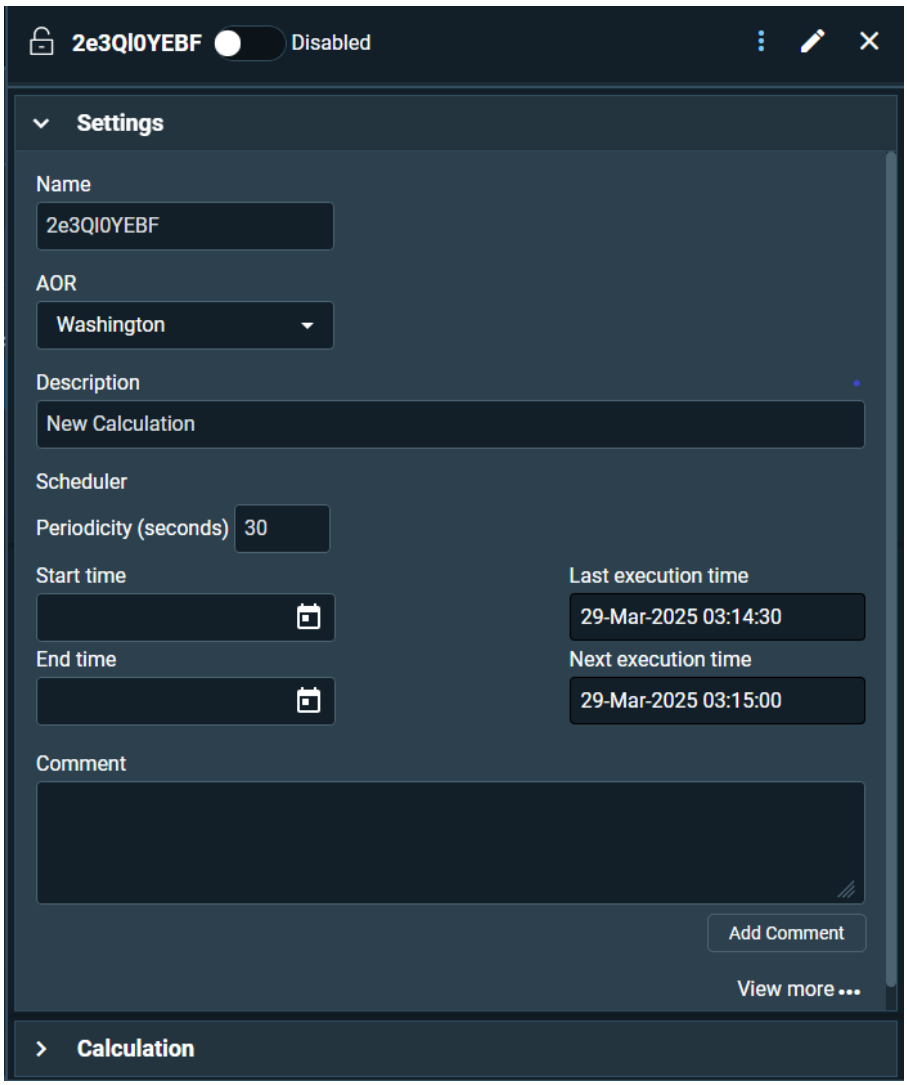
   - From the Actions menu, click **Add to a UDC**.

     The selected point is added as an output to the calculation.



- Navigate to the new calculation by returning to the **User Defined Calculations Summary** display and selecting the desired calculation.

3. Use the Settings panel to define general properties of the calculation.



> **Note**
>
> The **Name**, **Description**, and **Periodicity** fields are automatically populated with generated

> or default values during the creation of user-defined calculations. An AOR is assigned with the first value of the User Session AORs.

a. In the **Name** field, type a unique name that identifies the purpose of the user-defined calculation, and press Tab.
b. In the **AOR** field, select an appropriate AOR from the list. The list of AORs only includes those assigned to the user for that session.
c. In the **Description** field, enter the definition and purpose of the user-defined calculation, and press Tab.
d. In the **Periodicity** field, type a value for the number of seconds between each execution of the calculation, and press Tab.
e. (Optional) In the **Start time** field, select a value for the date/time that the calculation should begin execution, and press **Apply**.

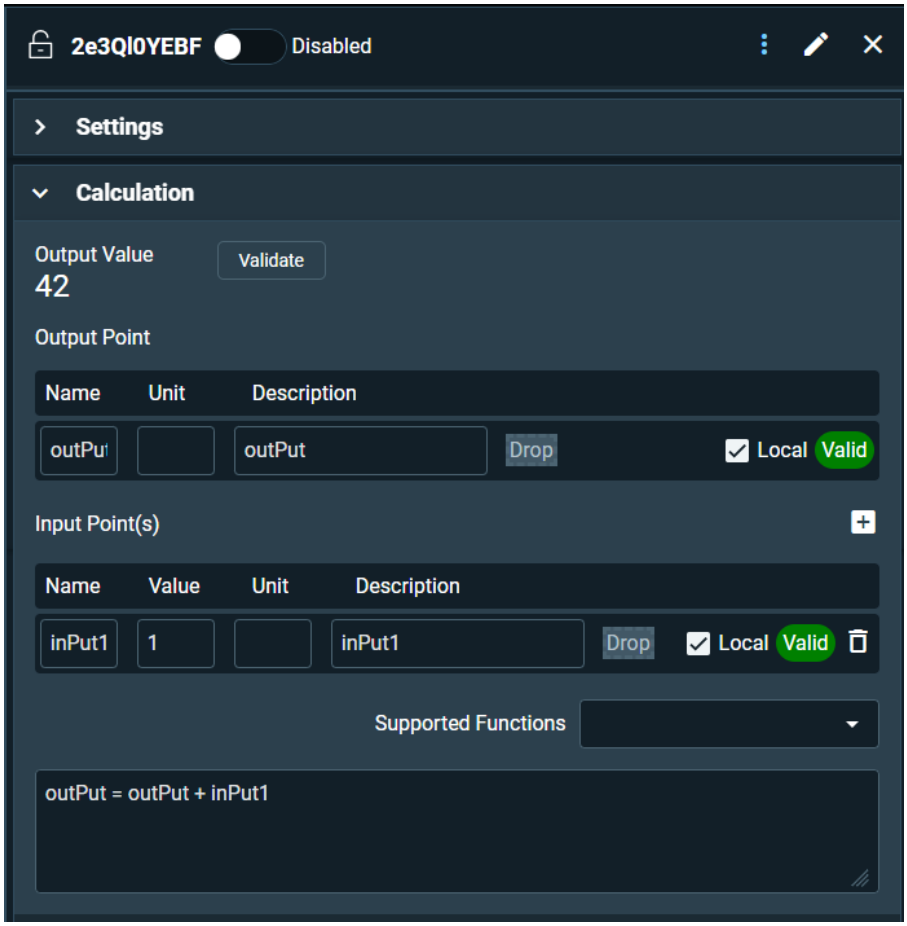
   If no value is provided, the calculation begins execution immediately upon being enabled.

f. (Optional) In the **End time** field, select a value for the date/time at which the calculation will automatically be disabled, and press **Apply**.

   If no value is entered, calculation execution is disabled only on operator request.

g. (Optional) In the **Comment** field, enter any relevant comments and press **Add Comment**.

4. Use the **Calculation** panel to define variables and outline the calculation definition.



> **Note**
>
> The first input/output record is shown by default. Output records cannot be deleted.

a. Additional input record(s) for each input to the calculation can be inserted using ⊞ symbol.

b. Input and output variables can either be local (that is, local to UDC database and not associated with SCADA records) or associated with SCADA Analog or Status Point records.

   You can associate Input/Output records by manually entering long names to identify SCADA records.

c. Repeat these steps for each Analog or Status Point record that you want to be an input to or output of the user-defined calculation.

d. In the definition (multi-line text) field, enter the calculation definition using input/output variables, and press Tab when done.

   For a list of valid general calculation operators, as well as examples of valid and invalid general calculations, see Additional User Calculation Information.

e. Once the calculation definition is added, click **Validate**.

   The calculation syntax is validated for valid inputs and outputs. If validation passes, the

calculation is marked **Disabled**.

If validation fails, an error message appears. Additional information about the error is provided in the **Log** display.

f. Correct any errors and revalidate.

g. When the function is validated, click the **Disabled** slide control, which will remove the ability to edit fields and enables calculation for execution. The slide control changes to **Enabled**.

h. The calculation service executes the newly enabled user-defined calculations. The **Next execution time** and **Last execution time** fields are updated each time the calculation is executed.

2.2. Modifying a Calculation

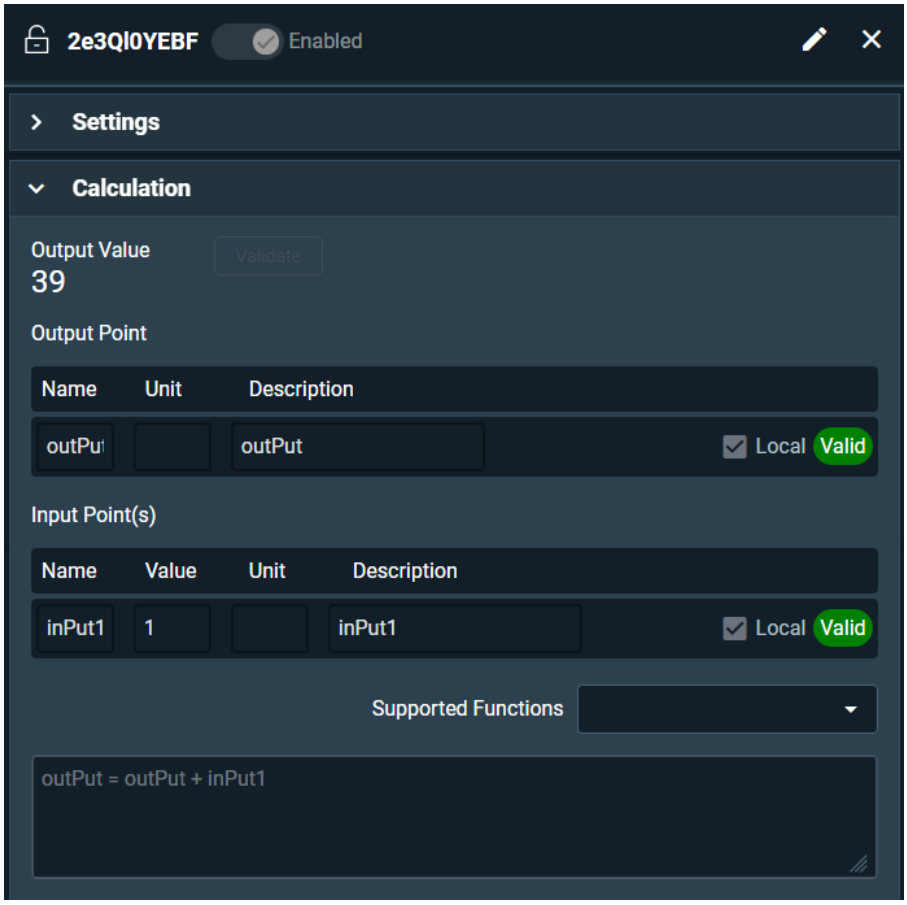
To modify an existing calculation:

1. Open the **User Defined Calculations Summary** display.

   The User Defined Calculations Summary appears, showing the available user-defined calculations.

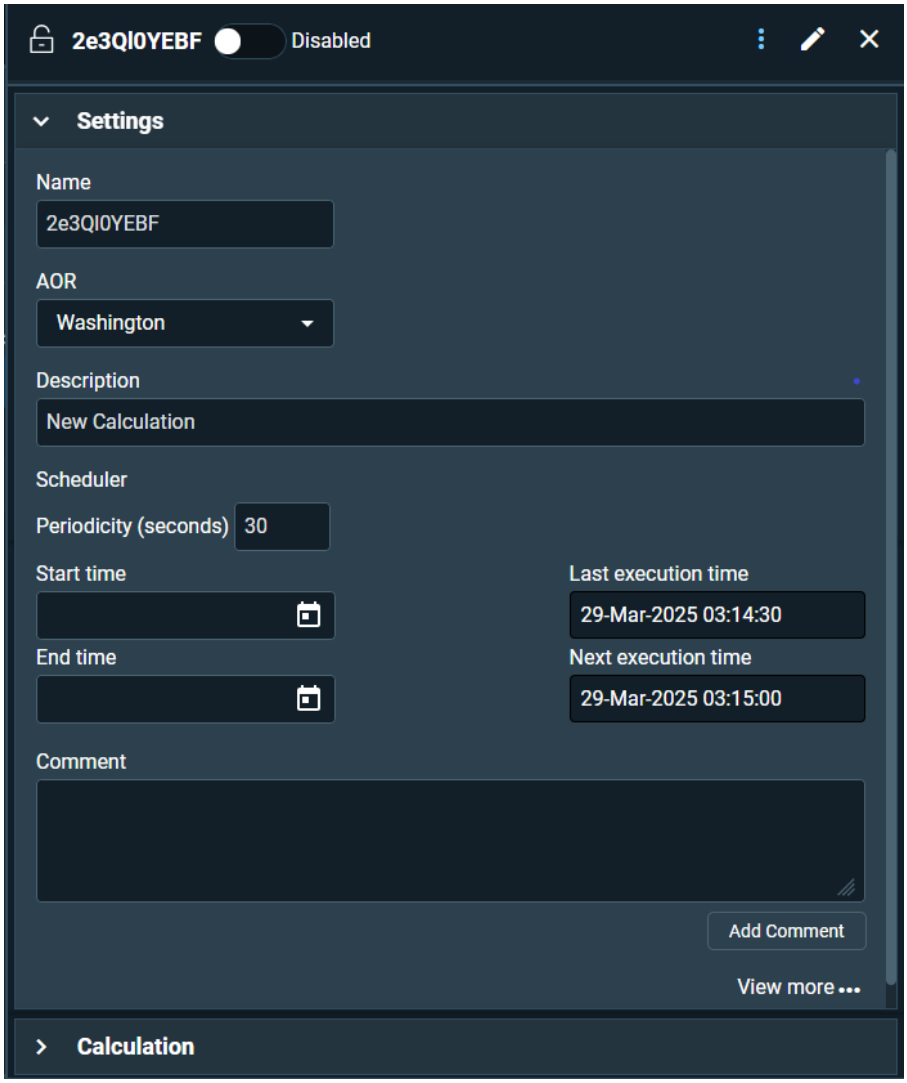
2. Click the name for one of the existing user-defined calculations.

   The User Defined Calculation display appears, showing data about the calculation you selected, in read-only mode.



3. Click the **Edit** icon to edit the calculation.

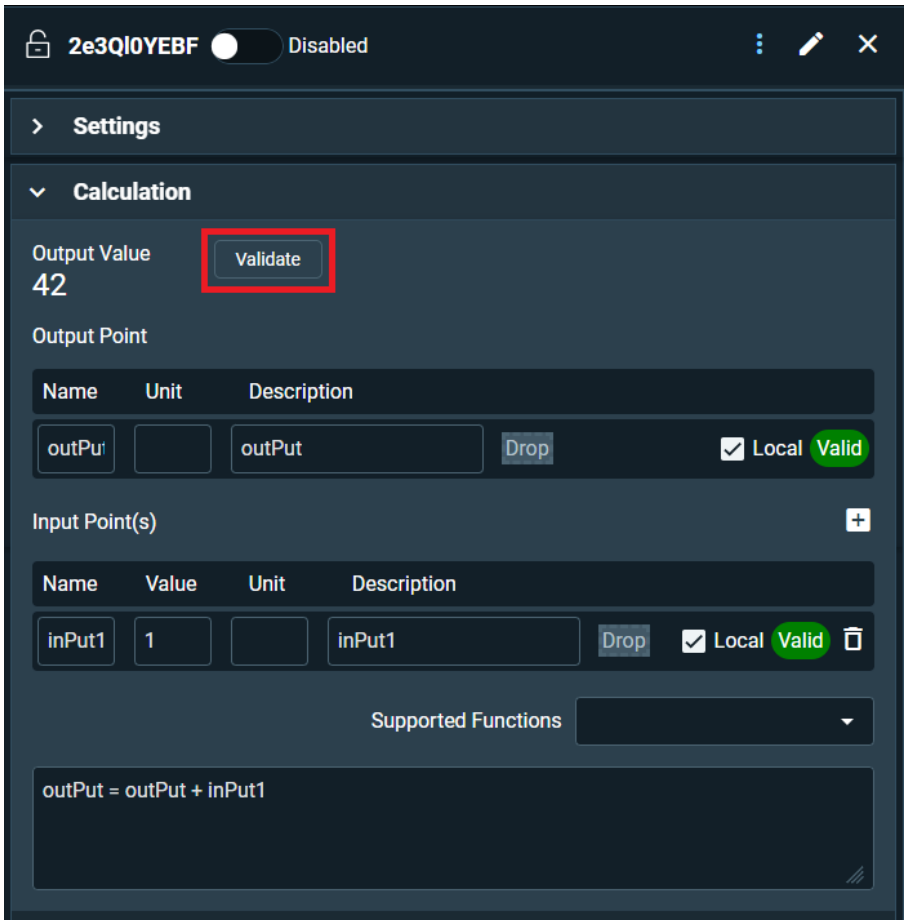
If the calculation is enabled, it will be disabled before any modifications are allowed. When the status changes to Disabled, the enterable fields that can be modified by the operator are displayed as enterable.



4. Select the appropriate fields, enter the required data for the calculation, and press Tab when done.

   When the new data has been accepted, the Calculation Status is set to **Uninitialized**.

5. When the calculation has been defined, click **Validate**.



The calculation is validated and marked as disabled.

6. Proceed as if a new calculation were being enabled.

2.3. Deleting a Calculation

To delete an obsolete calculation that is no longer required:

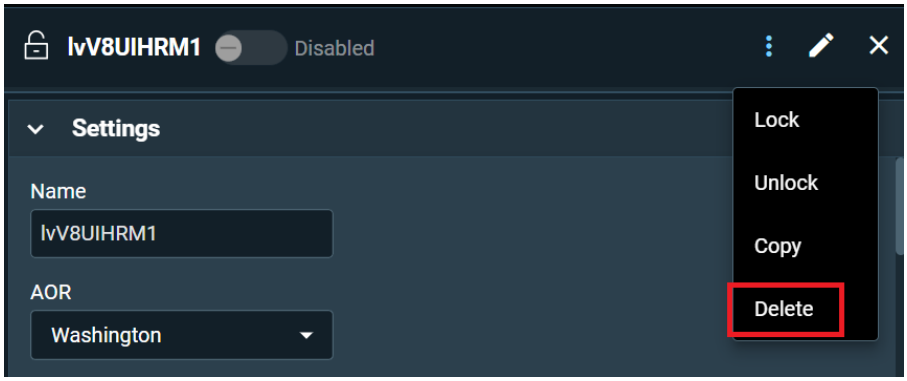
1. Open the **User Defined Calculations Summary** display.

   The User Defined Calculations Summary appears, listing the available user-defined calculations.

2. Click the name for one of the existing user-defined calculations.

   The User Defined Calculation display appears, showing data about the calculation you selected, in read-only mode.

3. Click the **Edit** icon to edit the calculation.
4. Click the menu icon to view available actions, and then click **Delete**.



The calculation is deleted from the database.

2.4. Locking and Unlocking a Calculation

Only the creator of the calculation or a system admin can lock or unlock a calculation.

View an existing calculation:

1. Open the **User Defined Calculations Summary** display.

   The User Defined Calculations Summary appears, listing the available user-defined calculations.

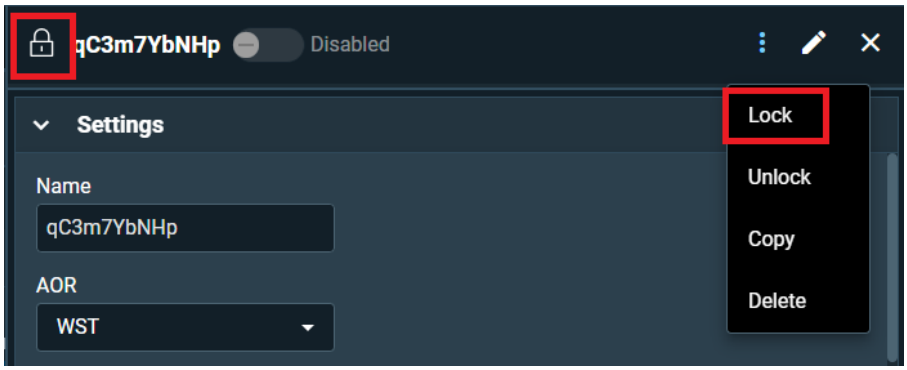
2. Click the name of one of the existing user-defined calculations.

   The User Defined Calculation display appears, showing data about the calculation you selected, in read-only mode.

3. Click the **Edit** icon to edit the calculation.

To lock the calculation:

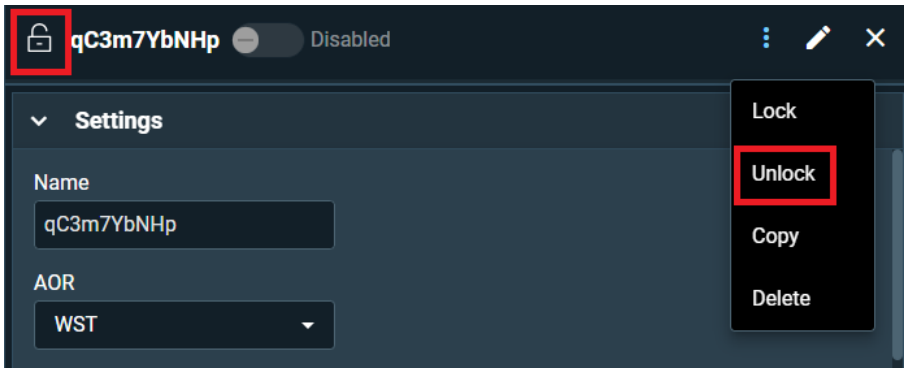
- Click the menu icon to view available actions, and then click **Lock**.



The calculation is locked, and only the creator or a system admin can edit it.

To unlock the calculation:

- Click the menu icon to view available actions, and then click **Unlock**.



The calculation is unlocked and edit is allowed by any user.

2.5. Copying a Calculation

To Copy an existing calculation:

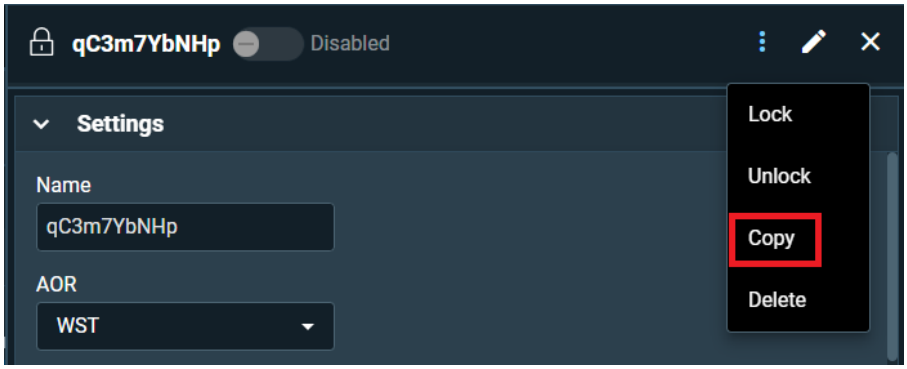
1. Open the **User Defined Calculations Summary** display.

   The User Defined Calculations Summary appears, listing the available user-defined calculations.

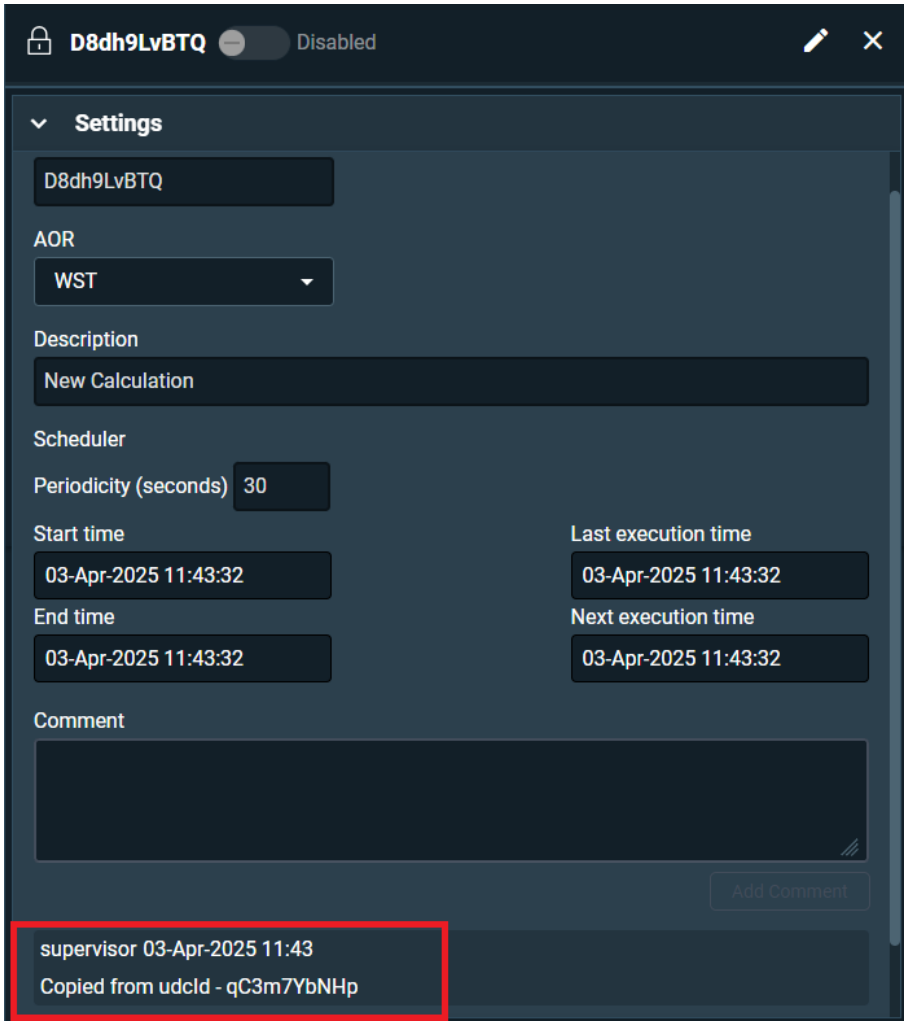
2. Click the name of one of the existing user-defined calculations.

   The User Defined Calculation display appears, showing data about the calculation you selected, in read-only mode.

3. Click the **Edit** icon to edit the calculation.
4. Click the menu icon to view available actions, and then click **Copy**.



A new calculation is created by duplicating content from the original calculation, except for the calculation name. A comment is added to indicate that it has been copied.



2.6. Pausing All Calculations

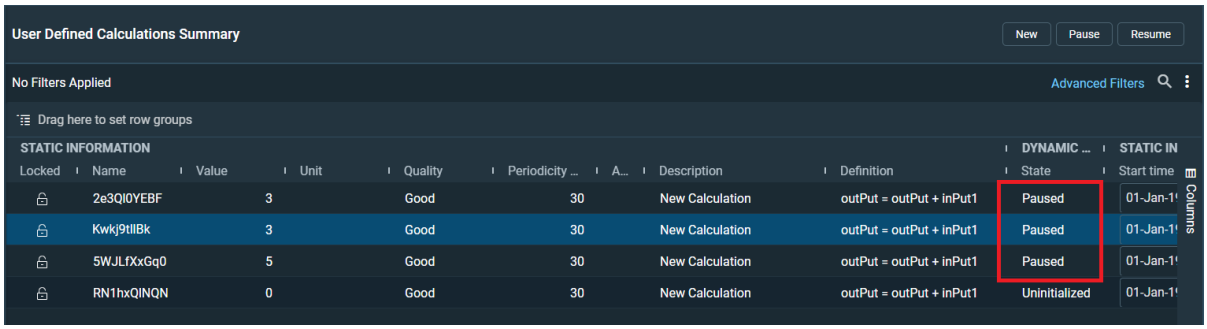
To stop execution of all enabled calculations simultaneously with a single action:

1. Open the **User Defined Calculations Summary** display.

   The User Defined Calculations Summary lists the available user-defined calculations.

2. Click **Pause**.

   All enabled calculations are paused and stop execution. The state appears as **Paused**, indicating to the operator that all enabled calculations have been paused.



2.7. Resuming All Calculations

This section describes the procedure for resuming execution of all enabled calculations after the Pause Calculations option has been selected.

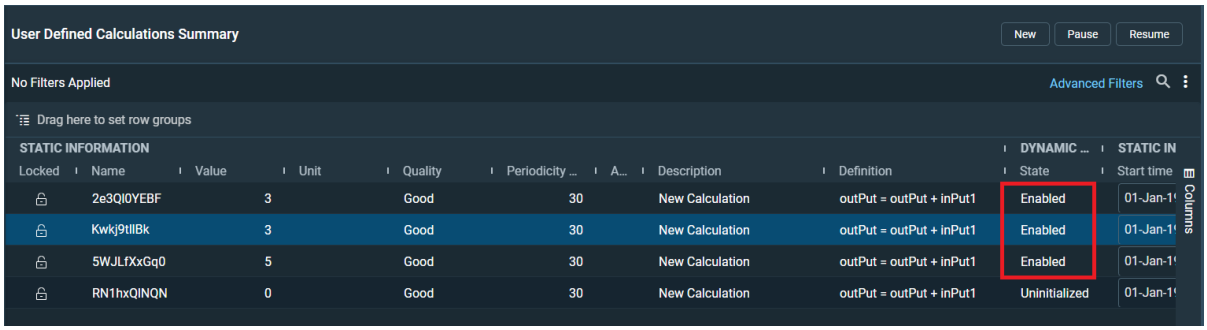
To resume all calculations:

1. Open the **User Defined Calculations Summary** display.

   The User Defined Calculations Summary lists the available user-defined calculations.

2. Click **Resume**.

   All paused calculations begin execution again as they were prior to the operator selecting the **Pause** option. The state again shows the text **Enabled**.



2.8. Additional User Calculation Information

The following sections contain a list of valid user calculation operators; examples of valid user calculations with arithmetic, Boolean, and logical operations; and examples of invalid user calculations with explanations as to why they are invalid.

2.8.1. Valid User Calculation Operators

The following table lists valid operators:

Table 1. Valid User Calculation Operators

Operator	Definition
+	Addition
-	Subtraction
/	Divide
*	Multiply
**	Exponentiation
.LT.	Less Than
.LE.	Less Than or Equal To
.GT.	Greater Than
.GE.	Greater Than or Equal To
.EQ.	Equal To
.NE.	Not Equal To
.OR.	Logical Or
.AND.	Logical And
IF-THEN	If Test, Then Expression
ELSE	Else Test
ELSEIF	ElseIf Test
ENDIF	End of IF-THEN or IF-THEN-ELSE block
IF-THENn	Embedded Test, Then Expression
ELSEn	Embedded Else Test
ELSEIF-THENn	Embedded ElseIf Test
ENDIFn	Embedded End of IF-THENn or IF-THENn-ELSEn block

The following table lists valid functions. Each function has one argument in the `form_func (arg)`:

Table 2. Valid User Calculation Functions

Function	Definition
_NOT	Logical Negation
_SQRT	Square Root
_LOG	Natural Logarithm
_LOGTEN	Base 10 Logarithm
_EXP	Exponential Function
_SIN	Sine
_SIND	Sine (degrees)

Function	Definition
_COS	Cosine
_COSD	Cosine (degrees)
_TAN	Tangent
_TAND	Tangent (degrees)
_ASIN	Arc Sine
_ASIND	Arc Sine (degrees)
_ACOS	Arc Cosine
_ACOSD	Arc Cosine (degrees)
_ATAN	Arc Tangent
_ATAND	Arc Tangent (degrees)
_SINH	Hyperbolic Sine
_COSH	Hyperbolic Cosine
_TANH	Hyperbolic Tangent
_ABS	Absolute Value
_INT	Truncation
_ROC	Rate of Change (between periodicity)
_ALARM	Raises an Alarm/Event. Takes 2 to 4 arguments. Arg 1: Variable on which alarm/event to be raised. Arg 2: User provided alarm/event text. Arg 3(optional): alarm/event priority. Arg 4(optional): alarm/event category. Example usage:- _ALARM(var1, "alarm text") _ALARM(var1, "alarm text", 3) _ALARM(var1, "alarm text", 2, "cat1")

Note

Expressions are evaluated from left to right without imposing any intrinsic order of operations based on the operations being performed.

However, parentheses can be inserted anywhere in the expression to specify the order in which expressions are evaluated. The number of left parentheses must match the number of right parentheses.

Tests for executing code blocks based on logical conditions must be entered in one of the following

formats:

```
IF (expression) THEN
  one or more statements
ENDIF
IF (expression) THEN
  one or more statements
ELSE
  one or more statements
ENDIF
IF (expression) THEN
  one or more statements
ELSEIF (expression) THEN
  one or more statements
ENDIF
IF (expression) THEN
  one or more statements
ELSEIF (expression) THEN
  one or more statements
ELSEIF (expression) THEN
  one or more statements
 .
 .
 .
ENDIF
IF (expression) THEN
  one or more statements
ELSEIF (expression) THEN
  one or more statements
ELSEIF (expression) THEN
  one or more statements
 .
 .
 .
ELSE
  one or more statements
ENDIF
```

Embedded IF-THEN-ELSE-ENDIF tests can be included as part of a higher-level IF-THEN-ELSE block, but they must be modeled with a special numeric identifier to identify them as lower-level IF-THEN test blocks.

The embedded IF statement must be paired with THENn, ELSEn, ELSEIF-THENn, and ENDIFn statements, where “n” is a numeric value between 0 and 9. There must be no spaces between the THEN, ELSE, ELSEIF, and ENDIF strings and the “n” numeric constant.

The following are examples of valid embedded IF-THEN-ELSE tests:

```
IF (expression) THEN
  IF (expression) THEN1
     one or more statements
  ELSE1
```

```
     one or more statements
  ENDIF1
  one or more statements
ENDIF
IF (expression) THEN
  IF (expression) THEN1
     one or more statements
  ELSE1
     one or more statements
  ENDIF1
ELSE
  IF (expression) THEN1
     one or more statements
  ELSEIF (expression) THEN1
     IF (expression) THEN2
       one or more statements
     ENDIF2
  ENDIF1
ENDIF
```

The value of an ANALOG variable is represented as a real number. The value of a POINT (Boolean) variable is represented as the integer 0 or 1.

2.8.2. Examples of Valid User-Defined Calculations

Example 1:

```
VARIABLE DEFINITION:
   KVIN  ANALOG IN -> STN   TYP   DEVICE   KV
   POUT  POINT OUT -> STN   TYP   DEVICE   STTS
FORMULA:
   IF (KVIN .LT. 110.) THEN  !If less than 110,
      POUT = 0 !  Set point OPEN
   ELSE !Else
      POUT = 1 !  Set point CLOSED
   ENDIF
```

Example 2:

```
VARIABLE DEFINITION:
   PNTIN POINT  IN  -> STN   TYP   DEVICE   STTS
   ANIN1 ANALOG IN  -> STN   TYP   DEVICE   SUM1
   ANIN2 ANALOG IN  -> STN   TYP   DEVICE   SUM2
   ANIN3 ANALOG LOC,  VALUE = 2
   ANOUT ANALOG OUT -> STN   TYP   DEVICE   AVG
FORMULA:
   !
   ! If input point set CLOSED, then do calc.
   !
   IF (PNTIN) THEN
```

```
      ANOUT = (ANIN1 + ANIN2) / ANIN3
   ENDIF
```

Example 3:

```
VARIABLE DEFINITION:
   B     ANALOG IN  -> STN   TYP   DEVICE   ALG1
   C     ANALOG IN  -> STN   TYP   DEVICE   ALG2
   A     ANALOG LOC -> STN   TYP   DEVICE   OUT
FORMULA:
   A = _SQRT(B) + C**3 !A = square root of
                       !    plus C cubed
```

Example 4:

```
VARIABLE DEFINITION:
   A     POINT IN  -> STN   TYP   DEVICE   PNT1
   C     POINT IN  -> STN   TYP   DEVICE   PNT2
   B     POINT OUT -> STN   TYP   DEVICE   PNT3
FORMULA:
   IF (A) THEN       !If A CLOSED, then
      B = _NOT( C ) !  Set B opposite of C
   ELSE !Else
      B = 1         !  Set B to CLOSED
   ENDIF
```

2.8.3. Examples of Invalid User-Defined Calculations

Example 1:

```
VARIABLE DEFINITION:
   B     ANALOG IN   -> STN   TYP   DEVICE   ALG1
   C     ANALOG IN   -> STN   TYP   DEVICE   ALG2
   D     ANALOG IN   -> STN   TYP   DEVICE   ALG3
   E     ANALOG IN   -> STN   TYP   DEVICE   ALG4
   F     ANALOG IN   -> STN   TYP   DEVICE   ALG5
   A     ANALOG OUT --> STN   TYP   DEVICE   OUT
FORMULA:
   A = ( (B * C / E) + SQRT(F) ) - 2) ** 3
REASON CALCULATION IS INVALID:
   1) The count of left and right parentheses do not
   match. More right parentheses than left
   parentheses. 2) The SQRT function is not preceded
   by an underscore (_).
```

Example 2:

```
VARIABLE DEFINITION:
   A     ANALOG IN   -> STN   TYP   DEVICE   ALG1
   B     ANALOG IN   -> STN   TYP   DEVICE   ALG2
```

```
   C     ANALOG OUT --> STN   TYP   DEVICE   ALG3
FORMULA:
   IF (A LT B) THEN
      C = A
   ENDIF
REASON CALCULATION IS INVALID:
   LT is not a valid operator, the operator
   must have the periods as part of the
   operator. It should be:
   IF (A .LT. B) THEN
      C = A
   ENDIF
```

Example 3:

```
VARIABLE DEFINITION:
   A     ANALOG IN   -> STN   TYP   DEVICE   ALG1
   B     ANALOG IN   -> STN   TYP   DEVICE   ALG2
   C     ANALOG OUT  -> STN   TYP   DEVICE   ALG3
FORMULA:
   IF (A LT B) THEN
      C = A
   ENDIF
REASON CALCULATION IS INVALID:
   LT is not a valid operator, the operator must have
   the periods as part of the operator. It should be:
   IF (A .LT. B) THEN
      C = A
   ENDIF
```

Example 4:

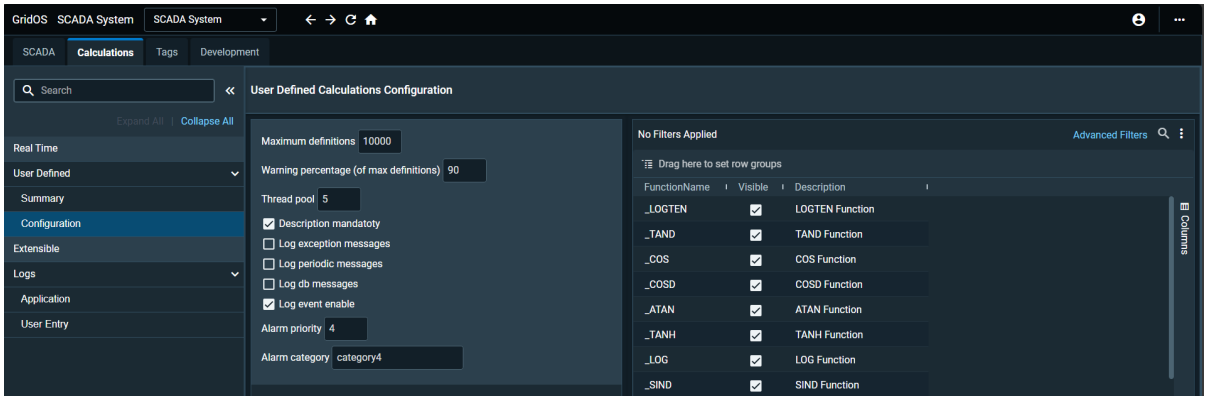
```
VARIABLE DEFINITION:
   A     ANALOG IN   -> STN   TYP   DEVICE   ALG1
   B     ANALOG IN   -> STN   TYP   DEVICE   ALG2
   C     ANALOG OUT  -> STN   TYP   DEVICE   ALG3
FORMULA:
   IF (A .LT. B) THEN
      IF (C .NE. B) THEN
         C = A
      ENDIF
   ENDIF
REASON CALCULATION IS INVALID:
   Embedded IF statements without the numeric to
   identify the embedded IF blocks. It should be:
   IF (A .LT. B) THEN
      IF (C .NE. B) THEN2
         C = A
      ENDIF2
   ENDIF
```

2.9. Configuration Displays

This section describes the calculation service level configuration.

1. Open the **User Defined Calculations Configuration** display.

   The User Defined Calculations Configuration appears, listing the default or previously modified settings of the calculation service.



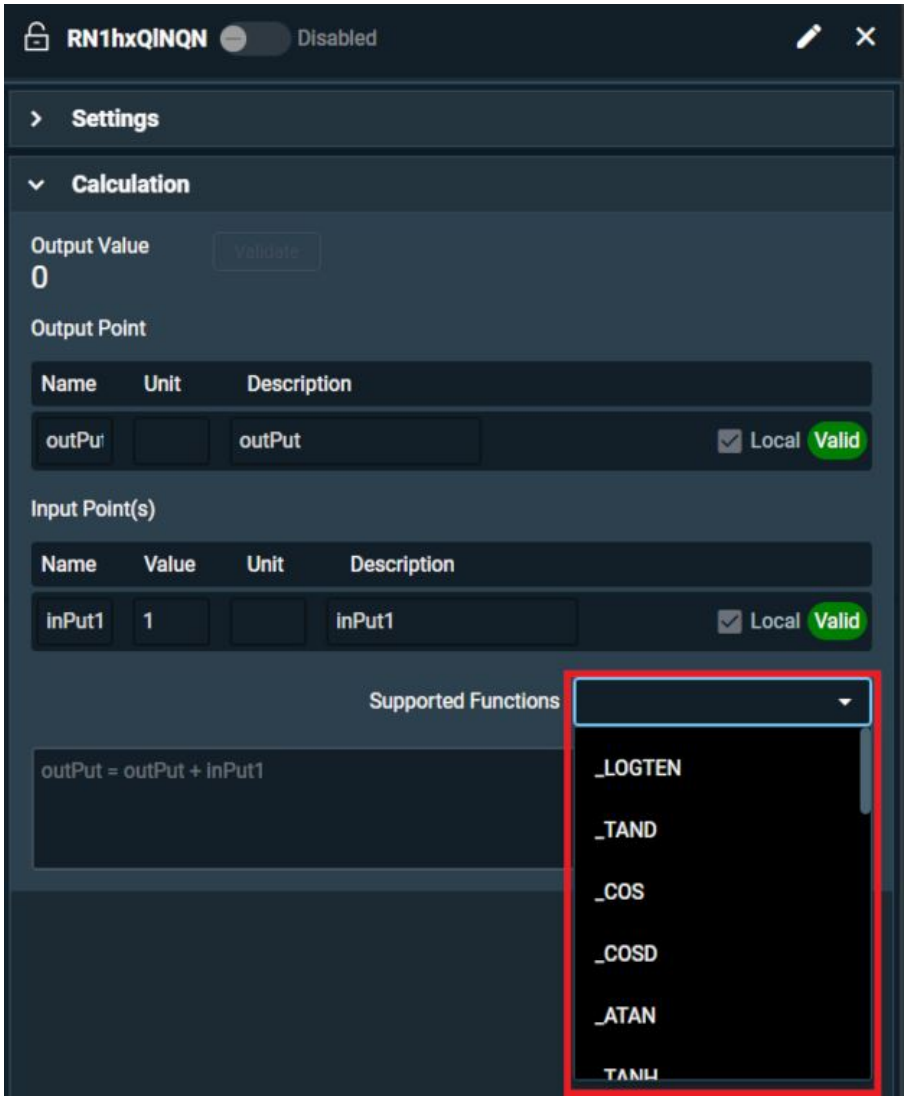
2. In the **Maximum definitions** field, enter a maximum number of user-defined calculations that can be defined, and press Tab.

   > **Caution**
   >
   > The solution has been tested with up to 10,000 calculations for optimal performance. Exceeding this limit may negatively affect the delivery solution's performance.

3. In the **Warning percentage** field, enter a percentage for the maximum number of user-defined calculations, and press Tab. A warning message will be logged once the number of defined calculations crosses this percentage.
4. The **Thread pool** field is intended only for use by GE Vernova service personnel.
5. Select the **Description mandatory** field to make sure the Description field is entered with valid text (including alphanumeric characters, arithmetic symbols, and `{ } ( ) ! = .` , symbols). Calculation validation fails when this property is selected.
6. Select the **Log exception messages** field to log detailed exception messages to the application log. This should be selected for a short periods of time to capture exception messages for debugging the calculation service in run-time.
7. Select the **Log periodic messages** field to log periodic execution messages to the application log. This should be selected for a short periods of time to capture extra log messages for debugging the calculation service in run-time.
8. Select the **Log db messages field** to log database method messages to the application log. This should be selected for short periods of time to capture extra log messages for debugging the calculation service in run-time.
9. Select the **Log event enable field** to send user entry messages as events to the Alarm/Event

Service.

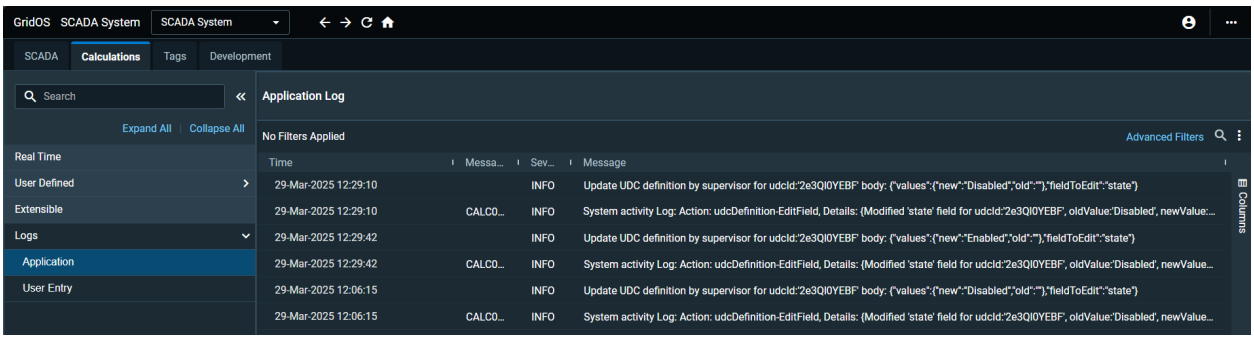
10. In the **Alarm priority** field, enter an alarm/event priority number (0-99), and press Tab. This priority is used as the default priority for the alarms/events generated by the calculation service.
11. In the **Alarm category** field, enter an alarm/event category, and press Tab. This category is used as the default priority for the alarms/events generated by the calculation service.
12. Select the **Visible** field from the functions table for a selected function. This property controls the list of function names that appear in the **Supported Functions** dropdown list within the **Calculation** panel.



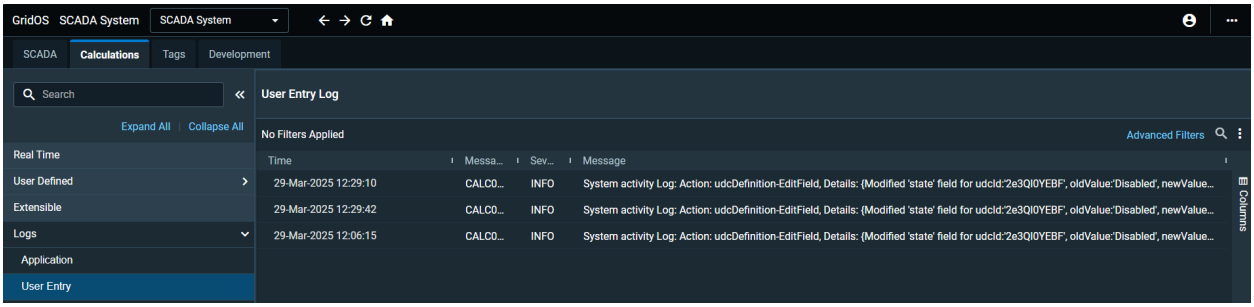
2.10. Log Displays

This section describes the **Application Log** and **User Entry Log** displays.

The **Application Log** display lists all application log messages emitted by the calculation service.



The **User Entry Log** display shows a pre-filtered list of user entry application logs emitted by the calculation service.



3. SCADA Controls

> **Important**
>
> This guide describes how to issue control commands from the Substation Tabular display. Currently, Open/Close operations are supported for 3-phase ganged and single-phase devices. Additional control types are planned for future releases.

3.1. Introduction

This section provides step-by-step instructions for operators to issue control commands from the Substation Tabular display. Controls allow authorized users to change the operating state of device and associated measurements within the power system.

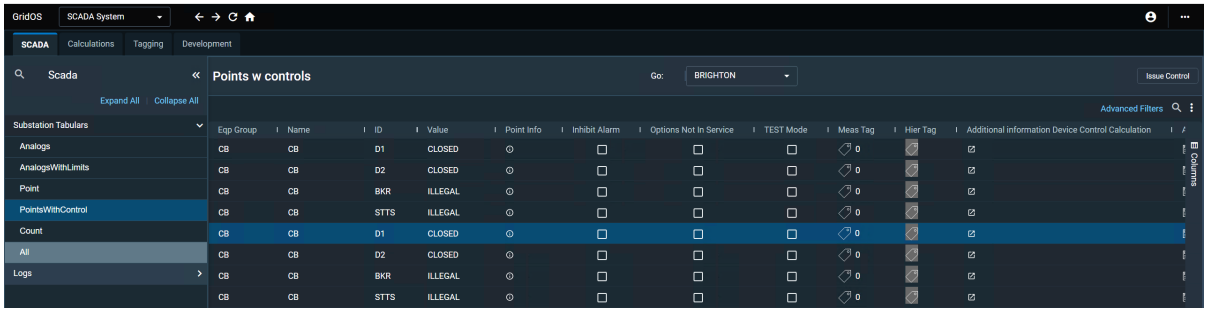
These instructions are intended for users with access to the GridOS interface and the necessary permissions to issue controls.

> **Note**
>
> This functionality is only available for a Hybrid solution consisting of GridOS and e-terra SCADA. In particular this is to address the needs of ADMS V1.

3.2. Issuing a Control Command

To issue a control command:

1. Navigate to the **Substation Tabular** display.
2. Select the **Points with Control** tab.



> **Note**
>
> The control side panel is currently available for all points under the **Points W Control** display. In a future release, it will be available only for controllable points.

3. Select the desired substation from the dropdown list. A tabular view displays the substation's

points, including the **Additional Information Device Control Calculation** column for control operations.

4. In the **Additional Information Device Control Calculation** column, click the control icon for the point you want to operate.

5. A side panel appears, showing point details and available actions. If a multiphase point is selected, the side panel displays current state for each phase. For single-phase points, only one control icon is shown.

   The fields in the side panel include:

   - **Alarm Indication**: Shows the current alarm status for the point. The icon is yellow if any alarms are active; otherwise, it is grey.
   - **Number of Tags**: Displays the count of tags applied to the point, indicating operational restrictions or notes.
   - **Area**: Indicates the operational area or zone for the point. Only one area is shown currently; multiple areas will be supported in a future release.
   - **Quality**: Reflects the current data quality or status (for example, Good, Bad, Uncertain). Only one value is shown currently; multiple values will be supported in a future release.
   - **Order ID**: Displays the switching order identifier for the control operation. This field is not editable and is currently populated with mock data. In a future release, it will be sourced from the Switching Application.

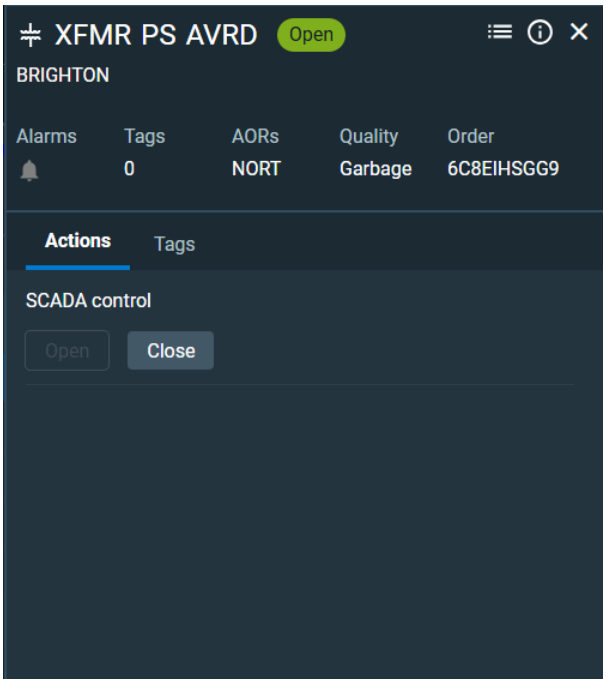


Figure 1. Single Phase Control Panel

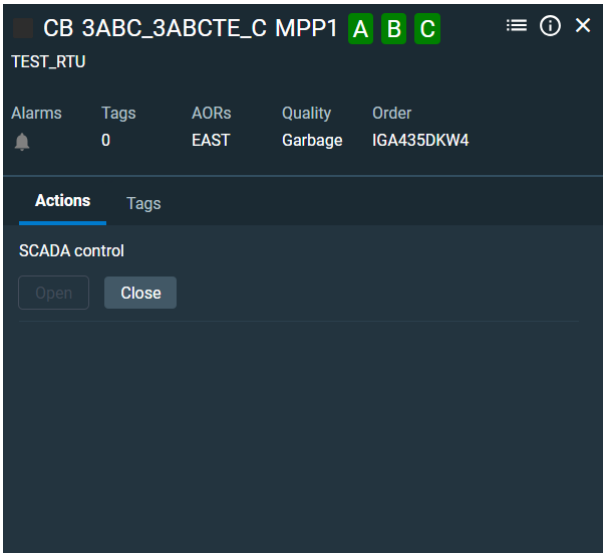


Figure 2. Multiphase Control Panel

6. Click the appropriate control operation (for example, Open, Close). Only valid operations for the current state are enabled. For example, if the point is Open, only Close is enabled.
7. A new side panel opens, presenting a **Check Before Operate** notification list. Review the notifications and click **Execute Control** to issue the command.

   Note the following:

   - Depending on the SCADA service configuration, the display of Check Before Operate (CBO) notifications may vary, as shown in the following images.

   > **Note**
   >
   > If the parameter `configuration.autocbo.enable` is set to `true` in the SCADA service configuration, then CBO validations are executed automatically and the resulting CBO notifications are displayed automatically when the control panel is opened.
   >
   > If the parameter is set to `false`, a button is displayed; upon clicking this button, the CBO validations are executed and the resulting CBO notifications are displayed in the panel.

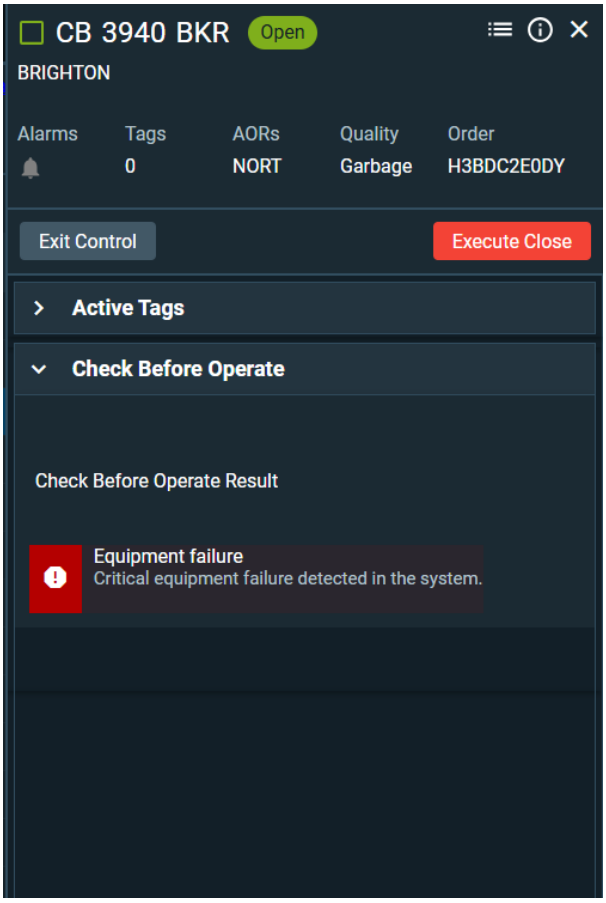


Figure 3. With autocbo Enabled

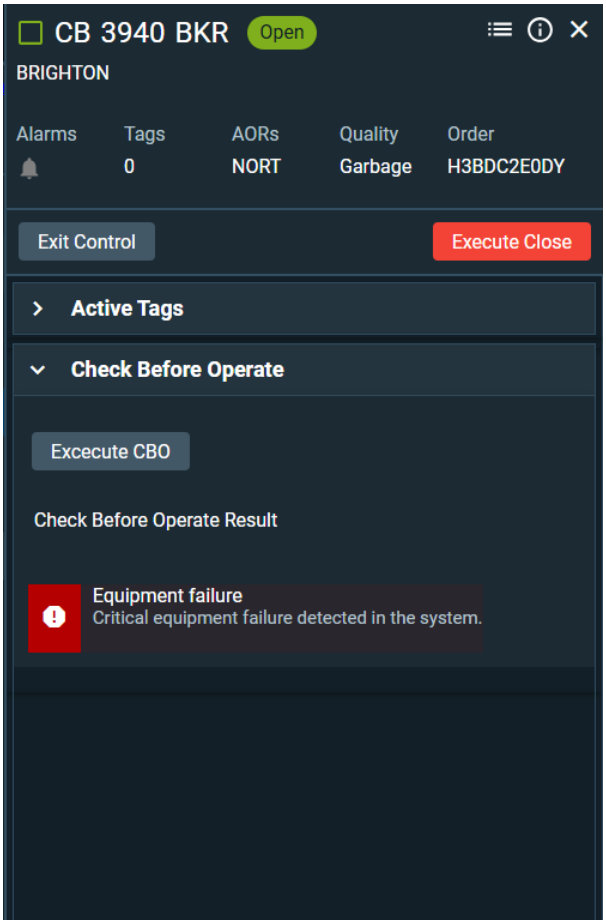


Figure 4. With autocbo Disabled and Button Clicked

> **Note**
>
> The **Check Before Operate** notifications are currently mocked for demonstration. In a future release, these will be sourced from ADMS or AEMS respectively depending on the deployment configuration.

- The selection panel for a multiphase point allows you to select and issue commands for individual phases, depending on device capabilities and permissions.

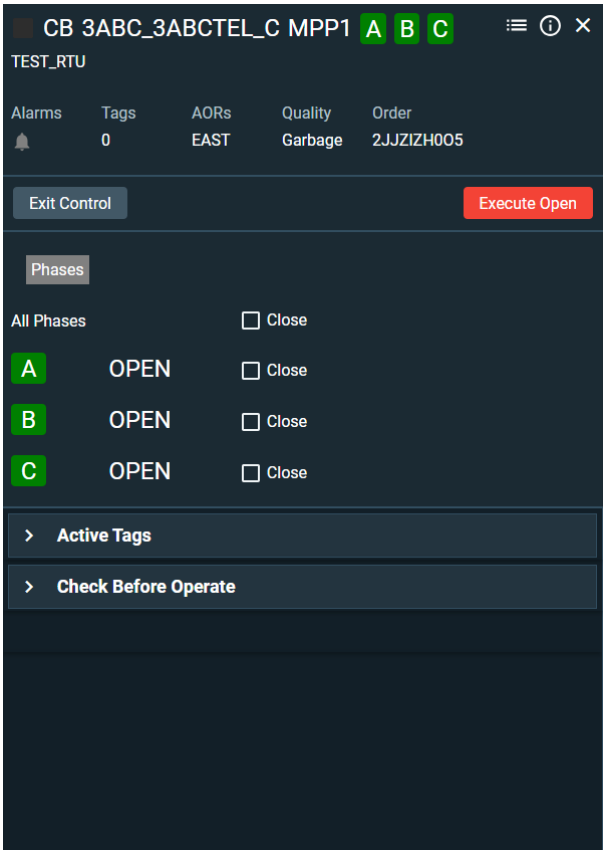


Figure 5. Multiphase Controls Selection Panel

> **Important**
>
> Even though the interface provides a multiple select checkbox, issuing multiple controls simultaneously will not work. The underlying Heritage SCADA system rejects control commands sent to the same point before a required WAIT time period has elapsed. Please issue controls one at a time for each point.
>
> The UI will be changed to a radio-button in a future release.

8. After clicking **Execute Control**, the command is sent to the Heritage SCADA (**e-terra***scada*) system. The system returns a success or error message. A success message confirms the control was issued; an error message provides details (for example, insufficient permissions).

> **Note**
>
> Currently, the status of the issued control is not displayed in the interface. This feature is planned for a future release.

3.3. Control Limitations and Constraints

- Only one control operation can be issued at a time for a given point.

- Control Groups cannot be created through this interface.
- Control issuance is subject to user permissions, and assigned AORs.
- Controls are managed on the Heritage SCADA side and may be subject to synchronization delays before appearing in GridOS displays.

3.4. Troubleshooting and Error Messages

- If a control issuance fails, ensure the HERITAGE_CONSOLE parameter matches with the active console on the Heritage system.
- Only applicable control types are shown. If a type is missing, check system configuration.
- If control responses do not appear immediately after issuance, allow time for synchronization.

4. SCADA Control Logging and Event Flow

This section explains how the GridOS logs and tracks all SCADA Control operations. The system records every action with detailed context and outcome, and generates an event for audit and monitoring.

You can access logging and eventing for all Control operations, if you have the necessary permissions.

This section also explains how the system extracts event properties, handles permission errors, posts events to the Alarm Router Service, the control logs and where you can view the applicable logs and events.

4.1. Introduction

GridOS logs all SCADA Control activities and generates events, which the system posts to the Alarm Router Service for traceability and compliance. As an Operator or Administrator, you can use these logs and events to audit Control actions, investigate issues, and ensure operational accountability.

4.2. What Gets Logged and Why

The following information is captured/logged for every Control operation (issue, update, fail):

- The type of operation (control type)
- The name of the equipment
- The user and the "device" (console, PC, tablet, etc.) from where the tagging operation is initiated.
- The outcome (success, failure, or permission error)

Log messages use templates defined in `messages/scada-plugin.json`:

ID	Severity	Format	Usage
SOP002	INFO	'%s' control issued successfully for '%s' from %s by %s	Successful control operation
SOP801	ERROR	Error: %s	Generic errors
SOP802	ERROR	'%s' control failed for '%s' from %s by %s: %s	Failed control operation (including permission errors)

4.2.1. How the System Uses Properties

- **Logging:** The system uses these properties as parameters for the log messages (SOP002, SOP801, SOP802), ensuring that each log entry contains full context about the operation.

- **Event Generation:** When generating a `GridAlarmEvent`, the system includes these properties in the event message and metadata, providing traceability for downstream systems.

4.2.2. Example

For a successful control operation, the system might set the following properties:

Property	Value
DeviceState	"CLOSE"
DeviceName	"SUBSTATION.BREAKER.123"
DeviceId	"123e4567-e89b-12d3-a456-426614174000"
UserName	"jdoe"
UserConsole	"ConsoleA"

The system uses these values to generate a log message like:

```
'CLOSE' control issued successfully for 'BRIGHTON.CB.3940.D2.BKR.123' from Organizer by super1
```

and includes them in the `GridAlarmEvent` sent to the Alarm Router Service.

4.3. Permission Handling

If a user attempts a control operation without sufficient permissions, it will be denied. The following takes place:

- Logs SOP802 with the message "Insufficient Permission to issue Control" and all relevant properties.
- Returns an error response to the user.
- Posts an error event to the Alarm Router Service.

Example log message for permission error

```
CLOSE control failed for 'BRIGHTON.CB.3940.D2.BKR.123' from Organizer by reader1: Insufficient
Permission to issue Control
```

4.4. Event Generation and Posting

After each operation (success or failure), the system prepares and posts an event to the Alarm Router Service. Each event includes:

- Source, type, and kind (SCADA plugin values)
- Event message (log message, e.g., SOP002, SOP801, SOP802)

- Equipment UUID
- Priority, genesis, and context (from configuration)
- Source time (current time)
- Data and event metadata

The event is serialized to JSON and posted to:

```
http://${env:ALARM_ROUTER_SERVICE_HOST}/api/v1/alarms
```

4.5. Viewing Logs and Events

User can view log messages and events in these locations:

4.5.1. SCADA Logs

- All log messages (SOP002, SOP801, SOP802) are written to the application logs
- Access logs via SCADA application logs

4.5.2. Alarm Router Service

- All Control events (including errors and permission denials) are posted to the Alarm Router Service
- Events can be forwarded, stored, or displayed in:
  - Monitoring dashboards (e.g., Grafana, Kibana, Splunk), if the customer has these capabilities avaiable and configured (these are not part of GridOS SCADA)
  - Alarm/event management UIs
  - Notification systems (for example, the alert manager via GridOS Foundation, if configured)

4.6. Example Logging Flow

1. A user requests a Control operation (e.g., CLOSE).
2. The system extracts relevant properties (state, device, user, console).
3. The system logs SOP002 for success, SOP802 for permission errors, or SOP801 for generic errors.
4. The system generates and posts a corresponding event to the Alarm Router Service.

Example success log

```
'CLOSE' control issued successfully for 'BRIGHTON.CB.3940.D2.BKR.123' from Organizer by super1
```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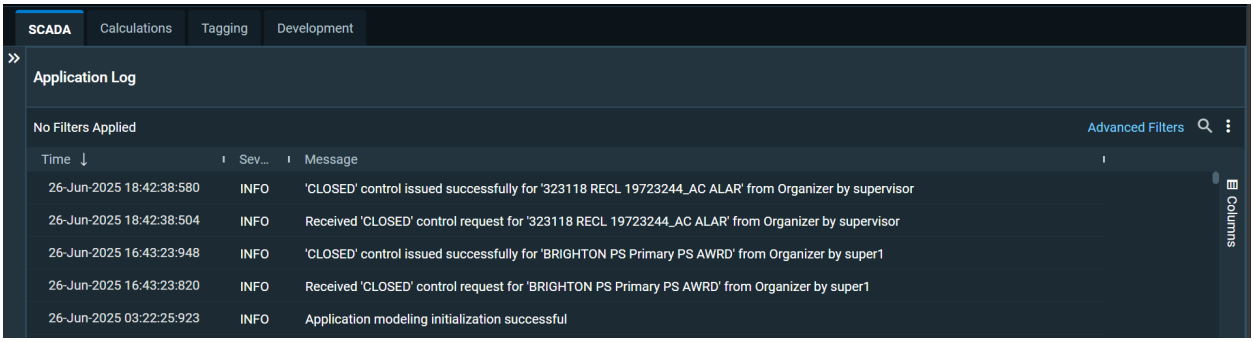
Example error log

```
CLOSE control failed for 'BRIGHTON.CB.3940.D2.BKR.123' from Organizer by reader1: Insufficient
Permission to issue Control
```

4.7. SCADA Application Log

Application log showing SCADA Control messages:



4.8. Summary

- All Control operations are logged with full context and outcome.
- Events are generated for every operation and posted to the Alarm Router Service.
- Logs and events are available in plugin logs, the Alarm Router Service, and downstream monitoring systems.

4.9. References

- `ScadaServiceRoutes.java`: src/main/java/com/ge/grid/organizer/plugin/scada/routes/ScadaServiceRoutes.java
- `messages/scada-plugin.json`: src/main/resources/messages/scada-plugin.json

5. SCADA Controls: Permissions and Area of Responsibility (AOR) Validation

This chapter explains how permissions and Area of Responsibility (AOR) validation work for SCADA control operations in the SCADA Organizer Plugin, including user roles, validation flow, and error handling.

5.1. Overview

SCADA control actions are tightly integrated with permissions and Area of Responsibility (AOR) controls.

- **Permissions**: Restricts which users can issue control commands.
- **Area of Responsibility (AOR)**: Restricts actions to specific device or operational areas.

Permissions and AORs together ensure that control actions can be done only by users authorized for that action and only on devices or operational areas that are authorized.

5.2. How Control Validation Works

5.2.1. Permissions Check

SCADA control actions are governed by the following user group permissions:

- **Writer Group**: Can issue control commands, such as changing device states or sending control requests.
- **Reader Group**: Can view control information and status, but cannot initiate or modify any control actions.

When you initiate a SCADA control action (for example, sending a command to change a device state), the system first checks the user group membership. If you do not have the required permissions, the action is blocked and a message is displayed (for example, "Insufficient Permission to issue Control").

5.2.2. Area of Responsibility (AOR) Validation

In addition to group permissions, SCADA control actions require validation of your Area of Responsibility (AOR):

- Your session is associated with one or more AORs, representing the devices or operational areas you are authorized to control.

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- When you initiate a control action, the system uses your session credentials (bearer token and client ID) to fetch your assigned AORs from a secure endpoint.
- The system compares the AORs specified in your control request with those assigned to your session.
- If your session includes at least one of the requested AORs, the control action is allowed; otherwise, it is blocked and a message explains the restriction.

AOR validation is performed immediately after the permissions check and before the control request is processed.

5.3. How It Works: Step-by-Step

The following example describes how permissions and AORs are checked when you complete a workflow:

1. Initiate Control: You submit a control request (such as changing a device state).
2. Permissions Check: The system verifies you are in the Writer group.
3. AOR Validation: The system fetches your assigned AORs and checks if you are authorized for the requested device or area.
4. Action Outcome: If both checks pass, your control action proceeds. If either check fails, you are notified with a message.

5.4. Summary

SCADA control actions are protected by both user group permissions and Area of Responsibility validation. You must be a member of the Writer group and have the appropriate AORs for the device or area you wish to control. Checks are performed before any control action is processed, and if you lack permissions or AORs, your request is blocked and you are notified with a clear message. This ensures that only authorized users can perform SCADA control operations, maintaining system security and compliance.

5.5. References

- SCADA Controls
- SCADA Control Logging and Event Flow

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6. Site Communication Status

6.1. Overview

The **Site Communication Status** display provides information about each site's communication status (a visual representation of the communication status between the SCADA system and remote sites), as well as options to control communication paths and view error statistics.

> **Important**
>
> As of this release, the control options are not functional and are read-only.

[site status display] | *gridos-scada/latest/user-guide/_images/site-status-display.png*

The table below lists all available columns in the **Site Communication Status** display, along with a brief description of each:

Column Name	Description	Shown By Default?
Name	Name of the site or entity	✓
Description	Description of the site or entity	✓
ID	Unique identifier	✗
Status	Status indicator (icon-based and color-coded) For more information, see Site Status.	✓
Status Description	Indicates the meaning of the status icon	✗
Not in Service	Indicates if the site is not in service	✓
Inhibit Alarms	Indicates if alarms are inhibited	✓
Disable Heartbeat	Indicates if heartbeat is disabled	✓
Block Suspect	Indicates if suspect blocking is enabled	✓
URL Call Up	Indicates if URL call up is enabled	✓
Point (Sent)	Number of points sent	✓
Analog (Sent)	Number of analogs sent	✓
Count (Sent)	Count of sent items	✓
Point (Received)	Number of points received	✓
Analog (Received)	Number of analogs received	✓
Count (Received)	Count of received items	✓

> **Note**
>
> The columns under **Sent** and **Received** show the number of items sent and received, respectively, since the system was last started or last failed.

6.2. Site Status

The table below provides details about the possible status values for a site, along with their descriptions and corresponding color-coded icons indicating the communication status of each site.

> **Tip**
>
> You can change the icons for each status by editing the table configuration file for the site status display and replacing the existing icons with the desired ones.

Status	Description	Default Icon
Local Site	The site is on the current server and requires no network connection.	
DTS Site	This site is fed by a DTS (Dispatcher Training Simulator) simulator in a DTS environment.	
Initializing	The SCADA process is trying to establish a connection with the remote site.	
Up	Communication with the remote site is functioning normally.	
Failed	The SCADA process has failed to establish a connection with the remote site.	
Not in Service	A user has removed the `InterSite` data path from service. The system will not attempt a connection.	

6.3. Site Node Details

When you expand a site, a table appears showing detailed information for its nodes.

[site status node details] | *gridos-scada/latest/user-guide/_images/site-status-node-details.png*

The table below lists all available columns in the node details table, along with a brief description of each:

Column Name	Description
Node ID	Unique identifier for the node
Site ID	Identifier for the site to which the node belongs
Node Mode	Indicates the mode of the node
Site State	Indicates the state of the site
Host Name	Name of the host machine for the node